Victor David Pérez De Los Ríos

MSc. in Artificial Intelligence | BSc. In Mechatronics | Medellín, Colombia

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Summary

Mechatronics engineer and MSc in Artificial Intelligence with a research background in visual computing, robotics perception, and autonomous systems. Two years as a research assistant at AVRIS (Visual Computing and Robotics group, Universidad EIA), with an undergraduate thesis on visual odometry for AGVs receiving an honorary mention, and a master's project on animal detection in dense aerial scenes. Subsequently spent five years deploying AI and intelligent automation solutions in production environments, developing a firsthand understanding of where autonomous decision-making breaks down when humans, uncertainty, and real business constraints collide. Seeking to return to research to address that problem at a principled level.

Refer to the GitHub Page https://victordpd19.github.io/ for quick extended abstracts in English of academic projects.

Education

MSc. in Artificial Intelligence (GPA 4.6/5.0) - Universidad de los Andes
Remote, Colombia | July 2023 – June 2025

- **AI Projects coverage:** Classic ML, EDA pipelines, MLflow, SoTA algorithms for neural networks (CNN, RNN, LSTM, VQGANs, Diffusion models, language and visual transformers), embeddings, LLM PeFT-LoRA finetuning.
- **Final master project:** Development of a deep learning model for counting and detecting animals in dense, occluded herds using aerial images **(**Final work in Spanish: https://l1nq.com/5yd7vjo **)**
- **Main tools:** Python, PyTorch, GCP, AWS, Railway, Postgres, Docker, Kubernetes.

BSc. Mechatronic Engineering (GPA 4.1/5.0) - Universidad EIA (Escuela de Ingeniería de Antioquia)
Medellín, Colombia | January 2016 – February 2021

- **Research:** Assistant for 2 years at research group AVRIS (Visual computing and robotics) resulting in:
    - Advanced standing of the subject: Computer Vision
    - Co-developed SIGTROV, an omnidirectional robot integrating Intel RealSense D435 RGB-D, ROS, PROFIBUS and Siemens PLC, as part of a concurrent industry internship at SAMCO Ingeniería.
- **Honorary mention in bachelor final project**: Visual odometry based-positioning system for Autonomous Guided Vehicles. (publication in Spanish: https://l1nq.com/tcbdmka)
- Colombian Government Excellence Undergraduate Scholarship (SPP2).
- Second Place Robotex Latam 2019: UiPath (digital bots) + physical line follower robots' competition.

Professional Experience

Employer: Lean Solutions Group (Lean Tech) – Artificial Intelligence Engineer
Medellín/Remote | December 2025 – Present
Clients served: Riverstone Logistics, RPM Freight

- **[Gen-AI]** Design, developed and test AI-powered based applications and automations for different US-based logistics clients using novel tools (n8n, Langfuse, Xano BaaS, Python + JS, Claude):
    - Design AI-voice agents and pathways for user experience surveys for after-deliveries and installation of General Electric Appliances, tracking user sentiment for decision making.
    - Improvement of backend "microapp" to visualize results of executions of AI workflows, resulting in better user experience and engagement to stakeholders.
    - Design and test multichannel Track and Trace AI agents that interact with logistic dispatchers and drivers to obtain pickup data via phone calls, SMS and emails.

Employer: Accelirate Inc. – Sr. RPA Automation Engineer
Medellín/Remote | January 2023 – December 2025
Clients served: Cenlar FSB, Morgan & Morgan, TPCi: The Pokémon Company International, Accelirate (internal)

- **[RPA + AI]** Engineered automation and analytics solutions for diverse US-based clientele in mortgage, legal, and media sectors utilizing the UiPath Automation Suite and using state of the art AI implementations, resulting in 20+ successful automation solutions and monitoring dashboards for their support.
- **[Gen-AI]** Conducted comprehensive research on enterprise-level applications of generative AI tools and executed proof-of-concepts (PoCs) with Python (Flask), typescript and Azure. Successfully completed these minimal viable artifacts:
    - Developed an AI-driven chat for responding to inquiries about candidates' CVs, HR, legal, and other documents using Retrieval Augmented Generation (RAG) techniques in Azure.
    - Constructed a LangChain Agent incorporating multiagent and tool usage capabilities.
    - Designed a UiPath bot capable of parsing unstructured emails in multiple languages and executing corresponding actions based on the content (a precursor of Communications Mining).

Employer: Neostella – RPA Team Lead
Medellín/Remote | September 2022 – December 2022
Clients served: Homebase, KnowBe4, Filevine

- **[Lead]** Led an RPA team distributed across Colombia and the Philippines of around 8 people, assisting on technical blockers as well as internal growth tasks,
- **[RPA]** Learned Robocorp RPA platform (Robot Framework + Python) and Workato API IPaaS for integration projects, obtaining most of the available certifications at the time, as well as multiple PoCs using these tools.

Employer: Neostella – UiPath RPA Developer
Medellín/Remote | June 2021 – September 2022
Clients served: Dinvy, Needham & Co, Vantage Point Title, Advantage Moulding, Neostella (internal)

- **[RPA]** Delivered and supported UiPath RPA processes for multiple US-based clients, reducing manual time consumed in ordinary tasks for the users, as well as increasing their performance and their positive perception of digital automation solutions.
- **[ML-Research]** Implemented an unsupervised machine learning model for document taxonomy clustering based on the DBSCAN model and OCR engines, embedded as a reusable custom C# activity library. The model was part of an internal research tool called "Pre-Doc". This project reduced the time consumed for document preparation step previous intelligent document extraction ML pipelines by around 20% for a single batch of around 1000+ documents.

Employer: SAMCO Ingeniería – Engineering Designer (Intern)
Medellín, On-Site | November 2019 – July 2020

- **[Mechatronic]** Assisted in multiple customized electromechanical design projects for local industries as well as programming Beckhoff industrial PCs (PLC + HMI) for an industrial AGV.

Technical Skills and Knowledge

- **ML and AI**: Statistics, SkLearn, PyTorch, OpenCV, NLTK, Huggingface models, OpenAI API, LLM Fine-tuning, LangChain, Nvidia Nemo.
- **Development tools**: Postman, Docker, Github, Jira, Notion, Confluence, VS Code, WinSCP, SOAP Ui.
- **Programming languages**: Python (Flask/FastAPI), SQL (Postgres), VB.NET, C#.
- **Cloud Services**:
    - Azure: AI 900 certified, Azure AI Foundry (Cognitive Services) and Azure ML, Blob storage, SQL DB, CosmosDB (NoSQL), Azure Functions.
    - AWS: S3, Lambdas, EC2, ECS, ECR
    - GCP: Vertex AI, Colab Enterprise, GKE, Cloud Runs, PubSub, CloudTasks, Cloud Runs.

- **Office tools**: MS Office, Google Workspace, LaTeX, Markdown,
- **RPA and integrations:** UiPath Studio and Cloud Orchestrator, n8n, API automation and Workato.

Soft Skills

- Assertive communication and active listening
- Diligence and decision making
- Leadership and empathy

Languages

- Native Spanish speaker
- Professional English proficiency (C1, TOEFL iBT 2026 – 110/120)
- Elementary German knowledge (A2)

References

- Carolina Cardona – Carolina Cardona Gómez | LinkedIn
  Product Specialist Engineer – Mechatronic engineer
  Scanreco AB - Stockholm, Sweden.
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- Luz Rivera, PhD - Luz María Rivera Rivera | LinkedIn
  Senior Business Analyst – PhD in Nanotechnology, Mechanical Engineer
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